

An Automated Digital Forensic Tool for Social Media Data Extraction and Evidence Documentation

1st Prince Raj

*Department of Artificial Intelligence and Machine Learning
Narula Institute of Technology
Kolkata, India
profile.princeraaj@gmail.com*

2nd Ankit Das

*Department of Artificial Intelligence and Machine Learning
Narula Institute of Technology
Kolkata, India
profile.ankitdas@gmail.com*

3rd Isha Saha

*Department of Artificial Intelligence and Machine Learning
Narula Institute of Technology
Kolkata, India
isha7saha@gmail.com*

4th Ayush Yadav

*Department of Artificial Intelligence and Machine Learning
Narula Institute of Technology
Kolkata, India
sidharth197502@gmail.com*

4th Alok Nath Pal

*Department of Artificial Intelligence and Machine Learning
Narula Institute of Technology
Kolkata, India
aloknath.pal@nit.ac.in*

Abstract—This document is a model and instructions for L^AT_EX. This and the IEEEtran.cls file define the components of your paper [title, text, heads, etc.]. *CRITICAL: Do Not Use Symbols, Special Characters, Footnotes, or Math in Paper Title or Abstract.

Index Terms—Panchnama, Electron.js, Google Firestore, PDFKit, Selenium WebDriver

I. INTRODUCTION

In terms of the digital world, social media platforms are now the go-to source for people, so much so that they've changed how we communicate and investigate crimes. Coming running over into the picture is a wealth of data generated by users that often contains the key to solving the most complex cases, zeroing in on potential suspects, and verifying witness statements.

Nowadays, law enforcement agencies are tapping into this digital goldmine, and using real-time information to unravel the inner workings of criminal behaviour and large-scale networks.

A. Challenges in Manual Evidence Collection

The process is riddled with challenges, when extracting information from social media. Coming heading through the tidal wave of social media data, one of the biggest hurdles is achieving a very high degree of accuracy, speed and consistency, and if this isn't met, critical evidence can be lost or altered because of the transitory nature of online content.

Well-known problems faced by investigators in the field of social media investigation include documentation errors, which arise when manually creating the Panchanamas, or evidence

documents, and account access issues, where social media accounts don't open on desktop computers even when the correct login credentials are used, forcing examiners to resort to using dummy phones.

The chain of custody in manual evidence capture is also a serious problem, mainly due to the practical difficulties of taking screenshots, recording timelines and transporting them without compromising the integrity of the chain.

B. Proposed Solution

The investigative bodies require tools that can efficiently process the vast amount of data, when trying to navigate the hurdles of investigating social media cases. The Social Media Parsing Tool is an automated framework, initially developed as a Windows application and powered by Electron.js frontend and Node.js backend, that addresses these problems. It systematically captures posts, messages, timelines and account details, and to ensure that it accurately replicates the flow of information, the tool relies on Selenium-WebDriver to mimic how humans browse the net, virtually eliminating the chance of human error.

The tool's number one goal is to protect the integrity of the data and so stores this information in Google Firestore, and automatically produces PDF reports with the help of PDFKit, ensuring that the data is court-ready and legal.

II. RESULTS AND DISCUSSION

The implementation of the Social Media Parsing Tool demonstrates a significant advancement over traditional manual forensic methods. This section evaluates the performance

of the tool based on efficiency, accuracy, and its ability to withstand legal examination.

A. Efficiency and Time Optimization

The primary result of the automated system is a drastic reduction in investigation time.

- **Automation vs. Manual Labour:** By automating the extraction of posts, messages, and friend lists, investigators are relieved from the tedious process of manual data collection.
- **Rapid Documentation:** The tool's ability to generate comprehensive PDF reports with embedded screenshots immediately after parsing ensures that case files are ready for review in a fraction of the time required by manual "Panchnama" creation.
- **Resource Allocation:** Reduced collection time allows forensic experts to shift their focus toward high-level data analysis and drawing valid conclusions rather than administrative data gathering.

B. Data Precision and Integrity

The tool addresses the critical challenge of human error in digital forensics.

- **Elimination of Human Error:** Automation ensures that vital evidence is not missed or inaccurately documented due to investigator fatigue or oversight.
- **Consistency:** By mimicking user interactions through Selenium, the tool provides a consistent method for capturing data across Whatsapp and Telegram.
- **Court Admissibility:** The structural design emphasizes data integrity, ensuring that the evidence gathered can withstand legal scrutiny and be accepted in court proceedings.

C. Security and Scalability

The integration of cloud-based storage provides a robust environment for sensitive data.

- **Encrypted Storage:** Utilizing Google Firestore ensures that all collected evidence is encrypted at rest and in transit, preventing unauthorized access.
- **Platform Resilience:** While initially optimized for Windows, the modular architecture proved capable of handling the frequent API and interface changes common on social media platforms through proactive monitoring and script updates.

D. Discussion of Challenges

A notable observation during testing was the "Desktop Access" issue, where certain accounts failed to open on desktop browsers even with correct credentials. The development of a secondary version for Android was identified as a necessary mitigation strategy to ensure the tool remains effective regardless of the hardware used by the examiner.

III. CONCLUSION AND FUTURE WORK

The proposed tool represents a significant advancement in digital forensics and detection technology. By automating the extraction and documentation of social media evidence, the tool allows investigators to prioritize data analysis and valid conclusions over the labour-intensive and error-prone process of manual data gathering.

A. Summary of Contributions

The tool effectively addresses the core challenges faced by law enforcement agencies today:

- **Automation:** By utilizing Selenium to mimic human interactions, the tool streamlines the "Panchnama" process and ensures real-time capture of relevant data.
- **Integrity:** The system minimizes human error and guarantees that gathered evidence can withstand legal examination.
- **Security:** Integration with Google Firestore provides a secure, encrypted environment for sensitive investigative data.

B. Future Work

To maintain its effectiveness and adapt to the evolving digital landscape, several enhancements are planned:

- **Platform Expansion:** Development of an Android version to address instances where accounts cannot be accessed via desktop browsers.
- **AI Integration:** Implementation of AI-based analysis to provide deeper investigative insights and visual data analytics.
- **Enhanced Security:** Introduction of two-factor authentication (2FA) and cloud-based collaboration tools for remote investigator access.
- **Multilingual Support:** Developing capabilities to parse and generate reports in various languages to meet diverse regional needs.

Ultimately, this tool is poised to become an indispensable component of the digital investigator's toolkit, significantly enhancing the speed, accuracy, and security of social media investigations.

IV. SYSTEM ARCHITECTURE AND DESIGN

The proposed system is architected as a modular, scalable, and secure desktop application designed to streamline the digital forensic process. The architecture decouples the user interface, automation logic, and data storage into distinct layers, ensuring that each component can operate independently while maintaining seamless integration. This modular design facilitates future expansion to other platforms, such as Android or Web, without requiring a complete overhaul of the core logic.

A. Architectural Overview

The system follows a tiered architecture comprising the Client Presentation Layer (Frontend), the Application Logic Layer (Back-end), the Automation Engine, and the Data Persistence Layer.

- **Frontend (User Interface):** Built using Electron.js, the frontend provides a user-friendly dashboard that allows investigators to initiate data extraction tasks, track progress in real-time, and view generated reports. The choice of Electron.js allows for the development of a robust Windows desktop application using standard web technologies (HTML, CSS, JavaScript) while ensuring the interface is intuitive for users without extensive technical expertise.
- **Back-end (Core Logic):** The application logic is powered by a Node.js runtime environment. This layer acts as the central controller, managing communications between the user interface, the automation engine, and the database. It is responsible for task scheduling, error logging, and processing the raw data extracted from social media feeds into structured formats suitable for reporting.
- **Automation Engine:** The core functionality of the tool relies on Selenium WebDriver, which is integrated to simulate human interactions with web browsers. This engine automates the process of logging in, navigating through profiles, and capturing screenshots, ensuring that the data extraction is both comprehensive and reliable.
- **Data Persistence & Storage:** Google Firestore, a cloud-based NoSQL database, acts as the primary repository for storing both the structured text extracted from social media feeds and the associated evidentiary screenshots. This technology was explicitly selected for its high-performance real-time synchronization capabilities, which establish a seamless data pipeline between the backend and the frontend dashboard. As the automation engine parses new data, Firestore instantaneously pushes these updates to the user interface, ensuring that investigators have immediate access to the most current findings without requiring manual page refreshes.

B. Component Design and Data Flow

The data flow within the system is designed to minimize human intervention and maximize data integrity. The process begins when an investigator initiates a task via the dashboard.

- **Task Initiation:** The front-end sends a request to the Node.js back-end.
- **Automation Execution:** The back-end triggers the Selenium WebDriver, which launches a browser instance to navigate the target social media platform.
- **Data Extraction:** The automation engine parses the DOM elements of the social media feed to extract text (posts, messages, timelines) and captures screenshots of relevant evidence.
- **Processing and Storage:** Extracted data is sent back to the Node.js back-end, where it is structured and immediately pushed to Google Firestore for secure storage.
- **Reporting:** Upon completion, the Reporting Module utilizes PDFKit to fetch the stored data and generate a comprehensive PDF report with embedded screenshots, formatted for legal admissibility.

TABLE I
TECHNOLOGY STACK SELECTION

Component	Technology	Justification
Frontend	Electron.js	Cross-platform compatibility and responsive UI design.
Backend	Node.js	Efficient handling of asynchronous tasks and I/O operations.
Automation	Selenium WebDriver	Precise simulation of user interactions for reliable data capture.
Database	Google Firestore	Real-time data handling, scalability, and robust encryption.
Reporting	PDFKit	Flexible generation of complex, well-formatted forensic reports.

C. Security and Error Handling Design

Given the sensitive nature of forensic data, security is deeply embedded into the system architecture to ensure both confidentiality and admissibility. The tool leverages the advanced security infrastructure of Google Firestore to implement strict access control policies, ensuring that only authorized personnel can view or interact with the collected evidence. To further safeguard the data against unauthorized interception or tampering, the system employs industry-standard encryption protocols that protect the data both while it is in transit between the client and the cloud, and while it is at rest within the database. This dual-layer security approach ensures that the tool complies with the rigorous standards required for managing confidential law enforcement data.

Furthermore, a robust error handling mechanism is implemented within the Node.js backend to maintain the reliability and robustness of the tool. This module is designed to proactively detect and log a wide range of operational anomalies, including network timeouts, page loading blocks, and failed login attempts. These detailed error logs serve a dual purpose: they provide immediate feedback for troubleshooting runtime issues, and they offer critical insights into platform-specific

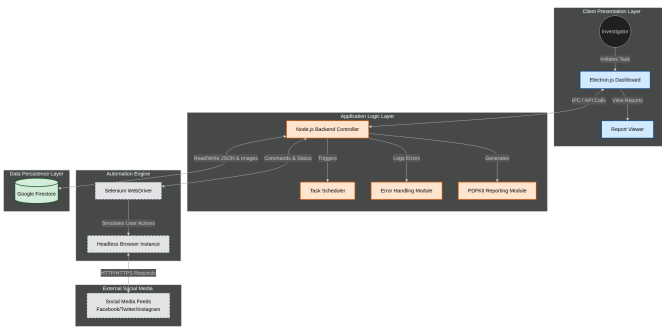


Fig. 1. High-Level System Architecture Diagram. The system separates the User Interface (Electron) from the Logic (Node.js) and Automation (Selenium) layers.

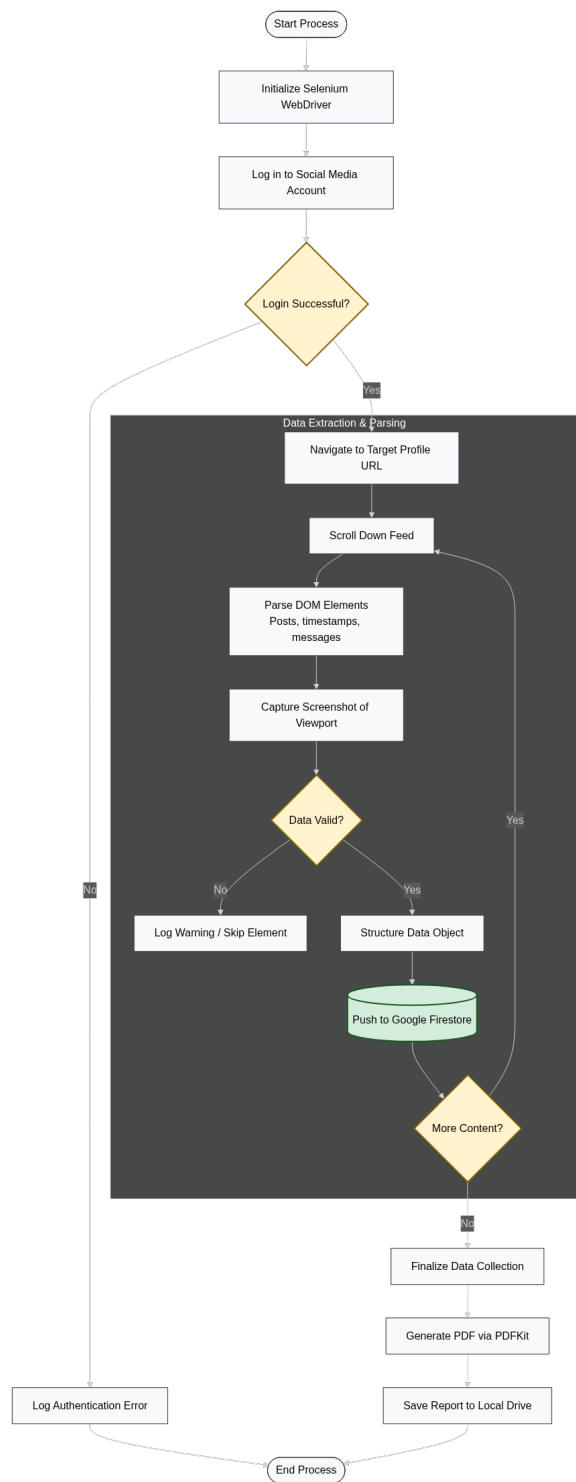


Fig. 2. Data Extraction Process Flowchart.

modifications, such as changes to DOM structures or APIs. By continuously monitoring these logs, the development team can rapidly identify compatibility issues and deploy updates, ensuring the tool remains resilient and operational despite the frequent interface changes inherent to social media platforms.

ACKNOWLEDGMENT

The preferred spelling of the word “acknowledgment” in America is without an “e” after the “g”. Avoid the stilted expression “one of us (R. B. G.) thanks . . .”. Instead, try “R. B. G. thanks. . .”. Put sponsor acknowledgments in the unnumbered footnote on the first page.

REFERENCES

- [1] G. Eason, B. Noble, and I. N. Sneddon, “On certain integrals of Lipschitz-Hankel type involving products of Bessel functions,” *Phil. Trans. Roy. Soc. London*, vol. A247, pp. 529–551, April 1955.
- [2] J. Clerk Maxwell, *A Treatise on Electricity and Magnetism*, 3rd ed., vol. 2. Oxford: Clarendon, 1892, pp.68–73.
- [3] I. S. Jacobs and C. P. Bean, “Fine particles, thin films and exchange anisotropy,” in *Magnetism*, vol. III, G. T. Rado and H. Suhl, Eds. New York: Academic, 1963, pp. 271–350.
- [4] K. Elissa, “Title of paper if known,” unpublished.
- [5] R. Nicole, “Title of paper with only first word capitalized,” *J. Name Stand. Abbrev.*, in press.
- [6] Y. Yorozu, M. Hirano, K. Oka, and Y. Tagawa, “Electron spectroscopy studies on magneto-optical media and plastic substrate interface,” *IEEE Transl. J. Magn. Japan*, vol. 2, pp. 740–741, August 1987 [Digests 9th Annual Conf. Magnetism Japan, p. 301, 1982].
- [7] M. Young, *The Technical Writer’s Handbook*. Mill Valley, CA: University Science, 1989.